

ARYAN RAJESH BURHADE

Nashik, Maharashtra | burhadearyan@gmail.com | +91 9226417910 | github.com/AB3117 | linkedin.com/in/aryanburhade-93427b295 | aryanburhade.me

AI & Data Science engineer who ships full-stack production systems and builds applied ML/agent tooling end-to-end. Has architected an LLM-driven natural-language-to-SQL agent, a computer-vision diagnostic pipeline with generative-AI reporting, and a validated satellite data-processing pipeline handling calibration, correction, and classification at scale. Comfortable owning a project from requirements through deployment: backend APIs, databases, model development, and client support.

SKILLS

Agentic AI & LLM: Agent Architecture (ReAct), Multi-step Reasoning, Tool-Calling, Autonomous Agents, LLM API Integration, Intent Parsing, Prompt Engineering, RAG, Embeddings, LangChain

Programming: Python, C, C++

Backend & Web: FastAPI, Django, React, Next.js, REST API Design, HTML/CSS

AI/ML & Computer Vision: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Image Classification, Object Detection, OCR, NumPy, Pandas

Data Engineering: Satellite/Geospatial Pipelines, Data Validation & QA, Clustering & Classification (K-Means, SAM)

Databases: PostgreSQL, MySQL, MongoDB

Cloud & DevOps: Git, Docker, Google Colab, Power BI, AWS (Cloud Practitioner)

EXPERIENCE

Freelance Software Developer & Systems Engineer

2024 - Present

- Designed, built, and deployed production backend systems for business clients end-to-end, from requirements through deployment and ongoing support: an AMC Management System, a Hospital Management System, a truck load-volume measurement system, and a niche community platform for graphologists.
- Systems are in active daily use by clients and have measurably improved operational efficiency and workflow.
- Delivered software troubleshooting and technical support for industrial clients.

PROJECTS

SpeakQL - Agentic Natural Language to SQL Engine

- 2nd Position, DBMS Project Competition
- Architected a ReAct-based agentic pipeline with tool-calling that connects to any SQL database via user-supplied credentials, maintains live schema awareness, and converts natural language into SQL: reasons over the request, selects tools, generates queries, and, after user approval, runs them directly against the connected database, including updates.

Automated Hyperspectral Data Processing Pipeline - Chandrayaan-2 IIRS

- 1st Prize, AICTE Innovation Cell 2025 | Selected, FOSS4G International Conference 2026 | 2nd Position, Project Based Learning Mini-Project Competition
- Built a complete Level-1 to Level-2 geospatial data pipeline for satellite hyperspectral imagery: radiometric/photometric calibration, physics-based thermal correction, and data-quality infrastructure (metadata parsing, per-pixel QA flagging, destriping) across a full orbital strip.
- Applied K-Means clustering and Spectral Angle Mapper classification, validated against RELAB reference spectra, to generate mineralogical surface maps; statistically validated a reliability issue in high-incidence-angle regions against the published ISRO/IIRS methodology.
- Currently building an agentic pipeline to fully automate the process end-to-end, taking raw hyperspectral data through calibration, correction, and classification to directly output mineral and surface maps.

DermaNova - AI-Based Dermatology Assistant

- Built a full-stack computer-vision diagnostic web app that classifies skin conditions from user-submitted images and layers in generative AI to produce personalized, report-based skincare recommendations.

AI Crop Disease Prediction & Recommendation System

- Built a computer-vision and environmental-data pipeline that predicts crop disease risk and recommends suitable crops from multi-source agricultural input.

WriteApp - Graphology Community Platform

writeapp.in

- Built a full-stack, thread-based community forum for graphologists with live annotation and tagging on handwriting samples, a gamification layer (XP, levels, badges) to drive engagement, and a standard blogs section for longer-form content.

CPAG - Course Selling Platform for Businesses

cpag.in

- Built a full-stack platform that lets a business sell its own courses online, with students browsing and consuming content organized by course, module, and lesson, and full payment integration for purchases.

EDUCATION

K. K. Wagh Institute of Engineering Education and Research

2023 - Present

- B.Tech in Artificial Intelligence & Data Science | SGPA: 8.45

H. P. T Arts and R. Y. K Science Junior College - Higher Secondary

2022 - 2023

St. Philomena Convent High School - Secondary

2020 - 2021

EXTRACURRICULAR, ACHIEVEMENTS & CERTIFICATIONS

- FOSS United Nashik - Technical & Design Lead, 2024 - 2026
- AWS Cloud Quest: Cloud Practitioner, Feb - Apr 2026